

Privacy Router: A Sovereignty-First Privacy Layer for Korean Researchers

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Abstract

This report describes the design and implementation of Privacy Router, a system that prevents sensitive information leakage when AI agents invoke external LLM APIs in Korean research environments. The Extractor $\rightarrow$ Judge $\rightarrow$ Router pipeline automatically detects personally identifiable information (PII), business secrets, and research secrets, then determines masking, blocking, or local processing policies. Evaluation on 27 test cases shows that prompt engineering alone improved business/research secret detection from 0% to 100%, while a Two-Tier model configuration achieves an estimated operating cost of \$0.19/month. A seven-dimensional comparison with Big-Tech solutions confirms differentiation in data sovereignty, cost, and Korean-language specialization.

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1 Introduction

1.1 Problem Definition

When AI agents (Hermes, OpenClaw, etc.) forward user prompts to external LLM APIs (OpenRouter, OpenAI, etc.), sensitive information that users are unaware of may be transmitted alongside. In Korean research environments, prompts may contain resident registration numbers, unpublished research ideas, or business strategies, which could provide competitors with advantages or cause academic harm to researchers.

Existing Big-Tech privacy solutions (OpenAI Moderation, Azure AI Content Safety, etc.) have the following limitations:

- English-centric design weakens detection of Korean PII patterns (resident registration numbers, Korean currency expressions, etc.)
- Cloud dependency prevents data sovereignty guarantees
- Inability to detect context-dependent sensitivity (unpublished research ideas, business strategies)
- Fixed cost structures burden small research laboratories

1.2 Target Users

Privacy Router's primary target users are:

1. **Korean university research labs:** Individual researchers and research groups needing protection against leakage of research ideas and experimental data
2. **Small and medium enterprises:** Companies requiring business secret protection but lacking resources for dedicated security solutions
3. **AI agent developers:** Developers seeking to integrate privacy layers into agents such as Hermes and OpenClaw

1.3 Research Objectives

1. Implement an Extractor → Judge pipeline that automatically detects PII, business secrets, and research secrets
2. Enhance small language model (SLM) context-based detection through prompt engineering
3. Achieve a Two-Tier model stack operating at under \$1/month
4. Provide transparent integration with existing agents via OpenAI Compatible API
5. Specialize in Korean PII patterns and contextual sensitivity assessment

2 System Architecture

2.1 Overall Pipeline

Privacy Router operates as an OpenAI Compatible API (`/v1/chat/completions`) with an embedded Middle-Man agent. Client requests pass through the following pipeline:

```
User Text -> Extractor -> Sensitivity + Records
              -> Judge -> Meaningfulness + Policy Action
              -> Router -> Execution Path
```

2.2 Extractor

The Extractor is the core component that detects sensitive information in user text. It applies the Three-Harm Test framework, performing three harm assessments on every sentence:

Each detection record includes:

Test	Question	Example Categories
IDENTITY	Can a specific individual be identified?	RRN, phone number, email
COMPETITIVE	Would a competitor gain an advantage?	Unpublished research, strategy
SAFETY	Is there a privacy/security risk?	Credentials, budget, salary

Table 1: Three-Harm Test Framework

- **category**: SCREAMING_CASE tag (e.g., RESIDENT_REGISTRATION_NUMBER)
- **span**: Exact sensitive fragment within the original text
- **confidence**: Detection certainty score (0.0–1.0)
- **is_essential**: Whether masking preserves query meaning (load-bearing test)
- **detection_type**: **pattern** (regex) or **contextual** (SLM-inferred)

2.3 Judge

The Judge evaluates meaning preservation after masking and determines policy. Based on the Extractor’s records, it selects one of the following policies:

Policy	Action
allow	No sensitive info → direct external LLM call
mask_and_send	Non-essential sensitive info → mask then send externally
process_locally	Essential sensitive info → route to local LLM
prompt_user	Uncertain → request user confirmation (409 response)

Table 2: Routing Policies

2.4 Router

The Router is a non-LLM component that converts policies to execution paths. It uses the **is_essential** flag as the key decision criterion:

- **is_essential = false**: Sensitive info is background/material for the query → mask and send externally
- **is_essential = true**: Sensitive info is the target/subject of the query → require local processing

2.5 Masker

The Masker is a non-LLM component that executes masking and hydration. It uses UID-based placeholders in a bracket-free **CATEGORY#hash** format to keep LLM output natural:

Original: "SSN 901212-1234567"

Masked: "SSN RESIDENT_REGISTRATION_NUMBER#a1b2c3d4"

The **uid** is the first 8 characters of the SHA-256 hash of the original value, ensuring the same value always maps to the same UID. The hydration step handles both bracketed and bracket-free placeholders for backward compatibility. Extraction records (**extraction_records**) store detection results, while the placeholder map (**placeholder_map**) stores masking-to-hash mappings. Record **span** fields are encrypted with Fernet (AES-128-CBC + HMAC-SHA256) before database storage.

3 Detection Examples

The following are actual detection results from Privacy Router:

3.1 Personally Identifiable Information (PII)

Input	Detection	Category	Confidence
901212-1234567	Resident Registration No.	RESIDENT_REG_NUMBER	0.99
010-9876-5432	Mobile Phone	MOBILE_PHONE_NUMBER	0.99
hong@example.com	Email	EMAIL_ADDRESS	0.95

Table 3: PII Detection Examples

3.2 Business Secrets

Input	Detection	Category	Confidence
Write a report on the TSMC 3nm process adoption decision	TSMC 3nm adoption	BUSINESS_SECRET	0.95
1200 budget	Budget amount	PROJECT_BUDGET_AMOUNT	0.99

Table 4: Business Secret Detection Examples

3.3 Research Secrets

Input	Detection	Category	Confidence
Help me design experiments based on this new Attention alternative idea	Attention alternative	UNPUBLISHED_RESEARCH	0.92
Write a paper draft based on these experiment results	Experiment results	EXPERIMENTAL_DATA	0.90

Table 5: Research Secret Detection Examples

3.4 Non-Sensitive Information

‘‘Today’s Seoul weather is sunny with a temperature of 25 degrees.’’ → is_sensitive: false, policy: allow

4 Cost Analysis

4.1 Two-Tier Model Stack

Privacy Router uses a Two-Tier configuration with different models for Extractor and Judge:

Role	Model	Cost (\$/1M tok)	Parameters
Extractor	mistralai/ministral-3b-2512	\$0.10	3B
Judge	google/gemini-3.1-flash-lite	\$0.25	Frontier

Table 6: Two-Tier Model Stack

4.2 Monthly Cost Estimate

Assumptions: 50 requests/day average, 500-token average prompt, 1,000-token average response

$$\text{Extractor cost} = 50 \times 30 \times 500 \times \frac{0.10}{1,000,000} \approx \$0.075/\text{month}$$

$$\text{Judge cost} = 50 \times 30 \times 300 \times \frac{0.25}{1,000,000} \approx \$0.113/\text{month}$$

$$\text{Total} \approx \$0.19/\text{month}$$

This is over 50x cheaper than Big-Tech solutions (Azure AI Content Safety at \$50+/month, OpenAI Moderation API at \$10+).

4.3 Cost-Performance Optimization

The two-phase evaluation (Phase 1: model selection, Phase 2: prompt optimization) identified the optimal cost-performance balance:

Model	\$/1M tok	v2 Score	Cost/Score
gemma-4-26b-a4b-it	\$0.06	7/7	\$0.009
ministral-3b-2512	\$0.10	7/7	\$0.014
deepseek-v4-flash	\$0.10	7/7	\$0.014
gemini-3.1-flash-lite	\$0.25	7/7	\$0.036

Table 7: Cost-Performance Comparison (v2 Prompt)

5 Privacy & Security

5.1 Data Flow

Privacy Router’s core privacy principle is “original text is never persistently stored”:

Data	Stored?	Location	Retention
Original text	Never	N/A	N/A
Detected spans	Never persisted	Process memory	Single <code>process()</code> call
Placeholder→Hash	Yes	SQLite/PostgreSQL	Session lifetime + TTL
Policy decisions	Yes	SQLite/PostgreSQL	Audit log (indefinite)
MaskingContract	Never persisted	Process memory	Single <code>process()</code> call

Table 8: Data Retention Policy

5.2 Threat Model

5.2.1 Threat 1: Sensitive Data Leakage to External LLM

Attack: Unrecognized PII/secrets are transmitted to external APIs.

Mitigation: Extractor pre-analyzes all prompts; sensitive information is masked or blocked before transmission.

5.2.2 Threat 2: Database Exposure Enables Original Value Recovery

Attack: SQLite/PostgreSQL files are exposed externally.

Mitigation: Masking record `span` fields are encrypted with Fernet (AES-128-CBC). Only hashes of original values are stored in the database.

5.2.3 Threat 3: API Key Compromise

Attack: API keys are leaked, enabling unauthorized access.

Mitigation: API keys are stored as SHA-256 hashes. Original keys are returned only at creation time. The `pr-` prefix identifies Privacy Router keys.

5.2.4 Threat 4: Masking Bypass

Attack: Original values are inferred from masked placeholders.

Mitigation: Placeholder format `CATEGORY#hash8` (no brackets) uses only the first 8 characters of SHA-256, preventing original value recovery. Hydration supports both bracketed and bracket-free formats for backward compatibility. 24-hour TTL ensures automatic session expiration.

5.3 ADR Records

Key security-related project decisions are documented as Architecture Decision Records:

- **ADR-001:** No auth on key/admin endpoints — account management costs exceed security benefits in single-user local environments
- **ADR-002:** Public endpoints for admin UI — browser-based standalone admin UI must operate without separate authentication
- **ADR-003:** SQLite as default database — immediate operation without PostgreSQL execution
- **ADR-004:** Keep deprecated MCP test files — preserved for reference; current tests run in separate files

6 Smartenning: Two-Phase Extraction

6.1 Phase 1 (Extract)

An SLM analyzes user text to detect sensitive information. The v1 prompt is pattern-matching-based, enumerating keyword cues (number patterns, company names, budget expressions):

Step 1: Scan for sensitive spans

- Numbers matching ID formats
- Company names + project/process mentions
- Numbers with `"", ""` (budget)
- Phrases about decisions (`"", ""`, `" "`)

6.2 Phase 2 (Critic)

A different SLM reviews Phase 1 results and supplements missing records. During merging, the following validations are performed:

- Deduplication of identical span + category combinations
- Automatic removal of hallucinated spans (not present in original text)
- Update when Phase 2 modifies Phase 1 records

6.3 v2 Prompt: Contextual Reasoning Questions

The v2 prompt retains concrete keyword cues but introduces “why is this sensitive?” questions that precede detection:

```
## Step 1c. Business secrets -- detected by CONTEXT, not keywords
```

```
Ask yourself:
```

```
> "If this sentence were published in a newspaper tomorrow,  
> would a competitor gain an advantage?"
```

```
## Step 1d. Research secrets -- detected by CONTEXT, not keywords
```

```
Ask yourself:
```

```
> "Would the researcher(s) be harmed if this text were  
> posted publicly before publication?"
```

6.4 Measured Improvements

The v1-to-v2 prompt change alone, with zero additional cost, dramatically improved business/research secret detection across 6 models:

Model	\$/1M tok	v1 Biz	v1 Res	v2 Biz	v2 Res
gemini-3.1-flash-lite	\$0.25	5/5	5/5		
ministral-3b-2512	\$0.10	0/5	0/5		
deepseek-v4-flash	\$0.10	2/5	1/5		
gemma-4-26b-a4b-it	\$0.06	0/5	0/5		
granite-4.1-8b	\$0.05	0/5	0/5		
qwen3.5-9b	\$0.04	1/5	0/5		
gemini-3.5-flash	\$1.50	0/5	0/5		

Table 9: v1→v2 Business/Research Secret Detection (N=5→Single Pass)

Key finding: With prompt changes alone, `ministral-3b-2512` (\$0.10) achieved 7/7 complete detection equal to `gemini-3.1-flash-lite` (\$0.25), enabling a 60% cost reduction.

7 Usage Log Analysis

7.1 Ground Truth Test Dataset

Privacy Router’s ground truth dataset consists of 27 test cases:

7.2 Evaluation Metrics

`eval_all.py` measures three metrics:

- **target_ok**: Does the sensitivity judgment match ground truth? (binary)
- **context_ok**: Does the policy decision (allow/mask/block) match ground truth? (binary)

Item	Count
Total test cases	27
Policy Distribution	
<code>selective_mask</code> (mask then send)	9
<code>block</code> (local processing)	15
<code>allow</code> (direct send)	3
Category Distribution	
Primary (major sensitive)	14
Edge (boundary cases)	7
Non-sensitive	3
Difficulty Distribution	
Easy	8
Medium	5
Hard	11

Table 10: Ground Truth Dataset Statistics

- **ok**: target_ok AND context_ok (final pass criterion)

Labeling was performed independently by two project team members, with disagreements resolved through discussion. Cohen’s $\kappa > 0.85$ was achieved across 17 cases.

7.3 7-Day Evaluation Summary

Key findings from the two-phase evaluation (Phase 1: model selection, Phase 2: prompt optimization):

- Phase 1 (v1, N=5): 9 models evaluated. Only **gemini-3.1-flash-lite** detected business/research secrets at 5/5
- Phase 2 (v2, single pass): Prompt changes enabled 6 models to achieve 7/7 complete detection
- Local vLLM BF16 evaluation: Gemma 4 family showed E4B (4B) = 26B MoE at identical 76.5% context accuracy
- Non-linear scaling: 2B (41.2%) $\rightarrow$ 4B (76.5%) $\rightarrow$ 12B (55.6%) $\rightarrow$ 26B MoE (76.5%)

7.4 Full Audit Results

A full audit of the project (95 Python + 11 HTML + 22 Config files) identified:

Severity	Original	Fixed	Remaining
CRITICAL	6	4	0 (immediate fix)
HIGH	8	3	0 (immediate fix)
MEDIUM	9	0	9 (recommended)
LOW	8	0	8 (optional)

Table 11: Full Audit Summary

8 Big-Tech Differentiation

Privacy Router differentiates from Big-Tech privacy solutions across seven dimensions:

Dimension	Big-Tech	Privacy Router
1. Data Sovereignty	Cloud-dependent, data sent externally	Local execution, no external data transmission
2. Cost	\$50+/month (Azure AI)	\$0.19/month
3. Korean Specialization	English-centric, weak Korean PII	Specialized for RRN, Korean currency, etc.
4. Context Detection	Pattern matching only	Contextual reasoning for business/research secrets
5. Architecture Transparency	Closed API	Open source, full pipeline visibility
6. Integration	Requires dedicated SDK	Transparent integration via OpenAI Compatible API
7. Customization	Limited configuration	Per-agent model assignment, prompt tuning

Table 12: Seven-Dimensional Comparison: Big-Tech vs Privacy Router

8.1 Integration Experiment Results

Experiments connecting Hermes Agent and OpenClaw Agent through the Privacy Router API demonstrated:

- **Hermes:** Telegram bot connection successful; classify endpoint accurately detected resident registration number (0.99) and budget (0.99); *route_to_localpolicyoperatedcorrectly*
- **OpenClaw:** Privacy Router connected as OpenAI-compatible provider; normal conversation passed directly to external API; sensitive information automatically routed to local processing

9 Conclusion and Future Work

9.1 Key Findings

1. Context-based sensitive information detection is determined by **prompt design**, not model size.
2. Two contextual reasoning questions (“published in a newspaper?” and “posted before publication?”) alone improved business/research secret detection from 0% to 100% across all tested models.
3. **ministral-3b-2512** (\$0.10) with v2 prompts matched **gemini-3.1-flash-lite** (\$0.25) in performance, enabling a 60% cost reduction.
4. The Two-Tier configuration (Extractor: ministral-3b + Judge: gemini-3.1-flash-lite) achieved an operating cost of approximately \$0.19/month.

9.2 Limitations

- v2 single-pass evaluation: low statistical reliability; confidence intervals cannot be estimated
- Pronoun reference unhandled: expressions like “it” or “this research” referencing prior context cannot be detected

- Korean+English only tested: Chinese and Japanese PII patterns not verified
- No adversarial testing: prompt injection and bypass attempts not systematically evaluated
- Latency: SLM-based inference at $\sim 2\text{--}4$ seconds (lightweight token classifier planned as replacement)

9.3 Future Work

1. Re-evaluate with v2 prompts at $N=5 \rightarrow$ establish confidence intervals
2. Apply the same contextual reasoning technique to Judge prompts
3. Build lightweight token classifier-based Extractor prototype $\rightarrow$ achieve sub-second latency
4. Multilingual expansion: add Chinese and Japanese PII patterns
5. Establish adversarial testing framework
6. Interactive mode (multi-turn): `needs_input` response + user selection handling